OFFICIAL TECHNICAL RELEASE • VERSION 2.4

DRABYASHREE NEPSE

Master Investor Encyclopedia & Quantitative Almanac

A comprehensive, mathematically verified compendium of the Nepal Stock Exchange (NEPSE) market microstructure: order flow mechanics, broker concentration metrics (BCR_3/BCR_5), 28 Japanese candlestick pattern recognition algorithms, interactive chart geometry, volatility frameworks, banking solvency directives (CAR/NPL), and capital preservation rules.

System: Drabyashree NEPSE Platform

Exchange: NEPSE NOTS Matching Engine

Architecture: Zero-Fabrication Verification

Date: October 2026

Drabyashree NEPSE — Master Investor Encyclopedia & Quantitative Almanac

The definitive guide to technical indicators, candlestick patterns, chart overlays, order flow microstructure, valuation models, regulatory directives, and execution frameworks for the Nepal Stock Exchange (NEPSE).

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0. Executive Glossary: Entry / Exit Analyzer Verdicts & Decision Taxonomy

The **Drabyashree Entry / Exit Analyzer** synthesizes price action, volume dry-up, broker accumulation metrics, historical analogs, and NEPSE-specific statutory constraints (circuit limits, T+2 lockups, and CGT friction) into actionable quantitative verdicts.

Verdicts operate within a strict **three-tier safety architecture**: 1. **Tier 1 — Hard Safety Vetoes**: Non-negotiable structural circuit, delivery, and institutional dumping halts that abort trades regardless of technical score. 2. **Tier 2 — Cautionary Overrides**: Secondary risk warnings (unfavorable RRR, transaction fee drag, festival liquidity drain, earnings lag) that require high conviction ($Score \geq 78$) to clear. 3. **Tier 3 — Score-Based Conviction Verdicts**: Mathematical ratings ($0–100$) evaluating setup strength once all safety gates pass.

0.1 Complete Verdict Reference Directory

Verdict Name	Category	Trigger Conditions	Tactical Action for Investor
NO TRADE (CIRCUIT CEILING TRAP)	Hard Veto	Price is within 2.5% of the $+10\%$ or $+15\%$ upper circuit ceiling.	**STRICTLY ABORT BUY.** Buying near upper circuit exposes trader to severe slippage and immediate T+2 reversal lock-in risk.
NO TRADE (T+2 DELIVERY EXHAUSTION TRAP)	Hard Veto	Day 3 or Day 4 of an unbacked vertical run where earlier buyers receive Demat delivery.	**AVOID BUYING.** T+2 settlement supply is hitting the market; institutional operators use retail momentum to exit.
NO TRADE (CRITICAL T+2 SETTLEMENT RISK)	Hard Veto	T+2 volatility score > 70; historical 2-day drawdown exceeds stop-loss room.	**STAY OUT.** Because NEPSE forbids selling on Day 0 and Day 1, expected drawdown will trigger loss before sale is legally permitted.
REDUCE / AVOID ENTRY (INSTITUTIONAL DUMPING)	Hard Veto	Top seller brokers (#58, #45, etc.) heavily offload while retail buys ($BCR_3\ Sell > BCR_3\ Buy$).	**EXIT / AVOID.** Smart money is liquidating holdings into retail euphoria; distribution pattern confirmed.
REDUCE / AVOID NEW ENTRY (BEAR STRUCTURE)	Hard Veto	Lower highs and lower lows beneath declining 50-EMA and 200-SMA.	**DO NOT AVERAGE DOWN.** Structure is in active Stage 4 decline. Preserve capital.
REDUCE / AVOID ENTRY (HYDRO DRY SEASON OVERHANG)	Hard Veto	RoR Hydropower scrip in dry season (Chaitra/Baisakh) with $60–70\%$ river discharge drop.	**WAIT FOR MONSOON.** Quarterly generation revenue collapse creates severe fundamental drag on price.
NO TRADE (UPSIDE < TRANSACTION FRICTION)	Caution	Distance to Target 1 is smaller than total round-trip fee (0.73%) + CGT (7.5%).	**PASS.** Even if target is hit, net realized return is zero or negative after statutory expenses.
NO TRADE (UNFAVORABLE RISK/REWARD)	Caution	Reward-to-Risk Ratio (RRR_1) is below $1.5:1.0$.	**WAIT FOR PULLBACK.** Risk taken outweighs potential upside. Wait for price to enter designated Pullback Zone.
NO TRADE (FESTIVE CASH DRAIN / LOW RVOL)	Caution	Pre-Dashain/Tihar festival liquidity drain; Relative Volume ($RVOL$) $< 0.70\times$.	**PRESERVE CASH.** Retail cash withdrawals drain market liquidity; breakout attempts have an 80% failure rate.
HOLD / AVOID NEW ENTRY (OPERATING LOSS)	Caution	Scrip reports negative EPS or operating losses in consecutive quarters.	**AVOID NEW POSITION.** Cannot distribute cash or bonus dividends under regulatory statutes.
REDUCE / AVOID NEW ENTRY (200 EMA RESISTANCE CEILING)	Caution	Major 200-session EMA sits directly overhead ($< 2.0\%$ away) as a thick supply shelf.	**WAIT FOR BREAKOUT.** Institutional supply overhang is heavy. Wait for decisive volume clearance above 200-EMA.
HOLD / AVOID CHASING (EXTREME VALUATION)	Caution	P/E or P/B multiple exceeds $2.5\times$ sector median without matching ROE growth.	**DO NOT CHASE.** Speculative multiple expansion leaves no margin of safety if sentiment cools.
HIGH-CONVICTION COIL (AWAITING TRIGGER)	Quantitative	$Setup\ Score \geq 82$, Pre-breakout consolidation under 20-day high with Volume Dry-Up.	**PRIME WATCHLIST.** Place buy stop at Breakout Trigger. Prepare for explosive markup impulse.
VERY STRONG SETUP	Quantitative	$Setup\ Score \geq 82$, active confluence of trend, broker accumulation, and $RRR \geq 2.0$.	**CONFIRMED BUY.** Full position allocation. Favorable risk/reward with structural stop protection.

Verdict Name	Category	Trigger Conditions	Tactical Action for Investor
COILED BASE (PRE-BREAKOUT RADAR)	Quantitative	$Setup\ Score\ 70–81$, base consolidating near 20-EMA / 20-day pivot.	**ACCUMULATE ON PULLBACK.** Accumulate in Dip Zone or enter on confirmed pivot break.
STRONG ENTRY ZONE	Quantitative	$Setup\ Score\ 70–81$, price is currently inside the validated low-risk Entry Zone.	**BUY IN ENTRY ZONE.** Clean swing setup with solid downside support floor.
BUY / ACCUMULATE	Quantitative	$Setup\ Score\ 58–69$, positive bias, healthy balance sheet, steady broker accumulation.	**GRADUAL ACCUMULATION.** Scale in with 50% initial tranche; add on confirmation.
HOLD / WAIT FOR CONFIRMATION	Quantitative	$Setup\ Score\ 45–57$, neutral technical posture; lacks volume conviction or trigger.	**HOLD EXISTING.** Do not initiate new position until volume conviction or breakout appears.
REDUCE / AVOID NEW ENTRY	Quantitative	$Setup\ Score\ 32–44$, deteriorating momentum, distribution signals, or breaking support.	**TIGHTEN STOPS.** Trim exposure on intraday bounces; do not initiate fresh longs.
EXIT / STAY OUT	Quantitative	$Setup\ Score < 32$, confirmed support breakdown, heavy institutional dumping.	**LIQUIDATE / EXIT.** Capital preservation mandatory. Sell into available market bids.

0.2 Mathematical Price Geometry: Why Entry and Breakout Prices Dynamic Change

Investors often ask: *Why does the Entry Price or Breakout Price recalculate between sessions?*

The system computes **Dynamic Execution Geometry**, not static price lines: 1. **LTP Anchoring & Volatility Calibration**: Entry zones are calibrated to the **Last Traded Price (LTP)** and the **14-day Average True Range (ATR)**. As market volatility expands or contracts, the clearance buffer dynamically adjusts:

$$\text{Clearance Buffer} = \max\Big(\text{SafeHigh}_{20} \times 0.0035,\; \text{ATR}_{14} \times 0.22\Big)$$

1. **Rolling 20-Session Swing High Pivot ($SafeHigh_{20}$)**: The breakout level is anchored to the highest high of the prior 20 trading sessions. As each trading day closes, older sessions roll out of the 20-day lookback window, shifting the pivot level.
2. **Pullback / Dip Zone Anchoring**: The optimal dip-buying zone is anchored to the **20-session Exponential Moving Average (EMA-20)** and recent 5-day base support, bounded securely above the structural stop loss:

   $$\text{Pullback Anchor} = \text{EMA}_{20} \quad \text{or} \quad \max\Big(\text{Structural Stop} \times 1.015,\; \text{LTP} \times 0.98\Big)$$

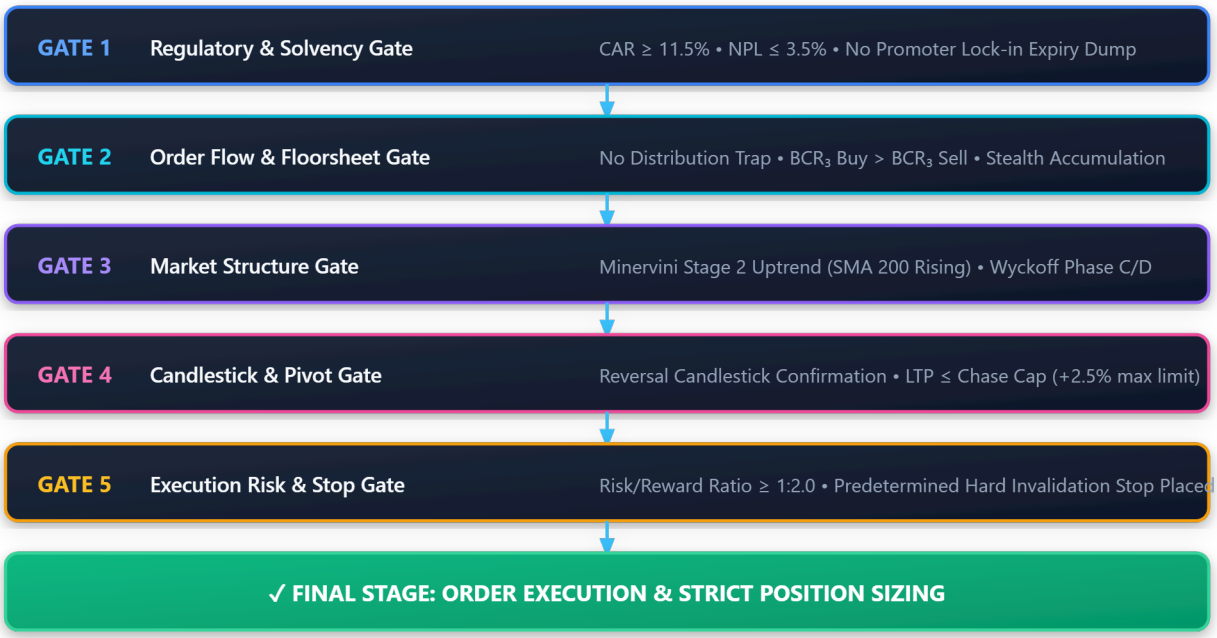
3. **Breakout Chase Cap**: To prevent retail traders from falling into the NEPSE T+2 delivery lockup trap, the engine enforces an invariant **Chase Cap** ($+2.5\%$ above the pivot). Buying above this cap is mathematically prohibited by the analyzer.

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1. Core Invariant Principles & Decision Hierarchy

Before deploying capital on NEPSE, every investor must understand that quantitative indicators do not exist in isolation. They operate within a strict **Precedence Hierarchy**:

FIGURE 1.1 — THE 6-GATE DECISION PRECEDENCE HIERARCHY



> ⭐ **IMPORTANT**
>
> **Hierarchy Rule**: An earlier gate always overrules a later gate. * If a stock displays a textbook Bullish Engulfing pattern on its chart (Gate 4), but the Floorsheet reveals active institutional dumping (Gate 2), the trade is **strictly aborted**. * If a stock looks fundamentally cheap with a high Margin of Safety (Gate 1), but has a non-performing loan ratio exceeding 5.0%, it cannot legally distribute dividends under NRB regulations and must be avoided.

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2. Services Hub & Flagship Decision Engines

2.1 Decision Probability Index (DPI: 0 to 100)

- **Definition**: A multi-factor Bayesian probability composite measuring the statistical likelihood of a stock generating positive net returns over the next 10 trading sessions.
- **Component Weights**:
- Volume & Microstructure Flow: 30%
- Trend & Momentum Alignment: 25%
- Value & Margin of Safety: 20%
- Macro Liquidity & Benchmark Trend: 25%
- **Interpretation**:
- $DPI \geq 75$: High mathematical conviction for swing capital.
- $DPI\ 50–74$: Neutral / Developing setup; requires secondary confirmation.

- $DPI < 45$: Negative statistical edge. Avoid opening long positions.

2.2 Minervini Stage 2 Trend Template + OAMI

- **Definition**: A systematic trend filter designed by US Investing Champion Mark Minervini, filtering for equities in confirmed institutional uptrends.
- **The 5 Invariant Stage 2 Filters**:
- Price $> 150\text{-Day SMA}$ and $> 200\text{-Day SMA}$.
- $150\text{-Day SMA} > 200\text{-Day SMA}$.
- 200-Day SMA sloping upward for at least 30 consecutive trading days.
- $50\text{-Day SMA} > 150\text{-Day SMA}$ and $> 200\text{-Day SMA}$.
- Current price must be at least $+30|\%$ above its 52-week low and within $25|\%$ of its 52-week high.
- **OAMI (Order Flow & Momentum Index)**: Confirms whether institutional accumulation volume is accelerating alongside price.

> **CAUTION**
> **Stage 4 Trap Warning**: Never attempt to "bottom fish" stocks trading below a declining 200-day moving average. In NEPSE, equities in Stage 4 (Markdown) frequently drift downward or consolidate without returns for 2 to 3 years.

2.3 VCP (Volatility Contraction Pattern)

- **Definition**: A chart geometry where a stock experiences progressively tighter price swings (e.g., Contraction 1: $-16|\% \ |rightarrow$ Contraction 2: $-8|\% \ |rightarrow$ Contraction 3: $-3|\%$) right beneath overhead resistance while daily turnover drops to multi-month lows.
- **Usage**: Proves that institutional supply has been absorbed. When buyers step in, price breaks out rapidly because no floating sellers remain.

2.4 8-Gate Multibagger Screener

- **Definition**: An institutional screening matrix requiring a stock to pass 8 simultaneous tests:
- **Clean Float**: Public float $\leq 40|\%$ of total shares.
- **Capital Efficiency**: Return on Equity (ROE) $> 15|\%$.
- **Earnings Velocity**: Positive EPS growth for 3 consecutive quarters.
- **Smart Money Footprint**: Top 3 brokers absorbing $\geq 40|\%$ of buy flow (BCR_3).
- **Prudent Leverage**: Debt-to-Equity ratio $< 1.5x$.
- **Trend Confirmation**: Price trading above 200-day SMA.
- **Lock-in Safety**: No promoter share lock-in expiration within the next 6 months.
- **Reasonable Valuation**: $PEG\ Ratio < 1.5x$.

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3. Smart Money, Floorsheet & Order Flow Microstructure

3.1 Broker Concentration Ratios (BCR_3 & BCR_5)

- **Definition**: The percentage share of total session volume bought or sold by the top 3 or top 5 brokerages:

$$\text{BCR}_3 = \frac{\text{Volume of Top 3 Brokers}}{\text{Total Session Traded Volume}} \times 100\%$$

- **Thresholds**:
- $BCR_3\ Buy \geq 40.0|\%$: Institutional cornering confirmed.
- $BCR_3\ Sell \geq 45.0|\%$: Institutional dumping warning.

> ⚠️ **WARNING**
> **The Asymmetry Principle**: A high $BCR_3\ Buy$ is **meaningless** if $BCR_3\ Sell$ is higher. If Broker #58 buys $40|\%$ of a stock, but Brokers #45, #34, and #28 collectively sell $55|\%$, the stock is experiencing **net institutional distribution**.

3.2 Distribution Trap (Pump & Dump Architecture)

- **Definition**: A deceptive condition where a stock's price is green ($+1|\%$ to $+6|\%$), but the floorsheet reveals that top brokers are dumping massive block quantities into retail market orders.
- **Identification Rules**:
- $BCR_3\ Sell \geq 45.0|\%$.
- $BCR_3\ Sell - BCR_3\ Buy \geq 10.0|\%$.
- Top 3 selling brokers are liquidating $> 20|\%$ of total session sell turnover.
- **Action**: When flagged 🚨 **DISTRIBUTION TRAP: DO NOT BUY**, fresh capital must be completely withheld.

3.3 Stealth / Quiet Accumulation

- **Definition**: Systematic institutional buying executed without causing rapid price spikes.
- **Markers**:
- Compressed daily price range ($\leq 1.5|\%$).
- Steady positive On-Balance Volume (OBV).
- High buy ticket size compared to sell ticket size.

3.4 Ticket Ratio (Trade Size Asymmetry)

- **Definition**:

$$\text{Ticket Ratio} = \frac{\text{Average Shares per Buy Contract}}{\text{Average Shares per Sell Contract}}$$

- **Interpretation**:
- $> 1.50x$: Institutional accumulation (Whales buying large 2,000–5,000 kitta blocks from retail selling 50–100 kitta lots).
- $< 0.70x$: Institutional offloading (Whales dumping large blocks into small retail bids).

3.5 Bilateral Cross-Trade Neutralization

- **Definition**: Matched transactions where the same broker ID is on both the buy and sell side of a contract (e.g., Broker #58 selling 50,000 kitta to Broker #58).
- **Usage**: The app neutralizes self-matched trades to prevent artificial inflation of accumulation scores. These represent promoter transfers or internal rebalancing, not open-market demand.

3.6 5-Depth Order Book & OBIR (Order Book Imbalance Ratio)

- **Definition**: Real-time aggregation of top 5 buy bids and top 5 sell asks:

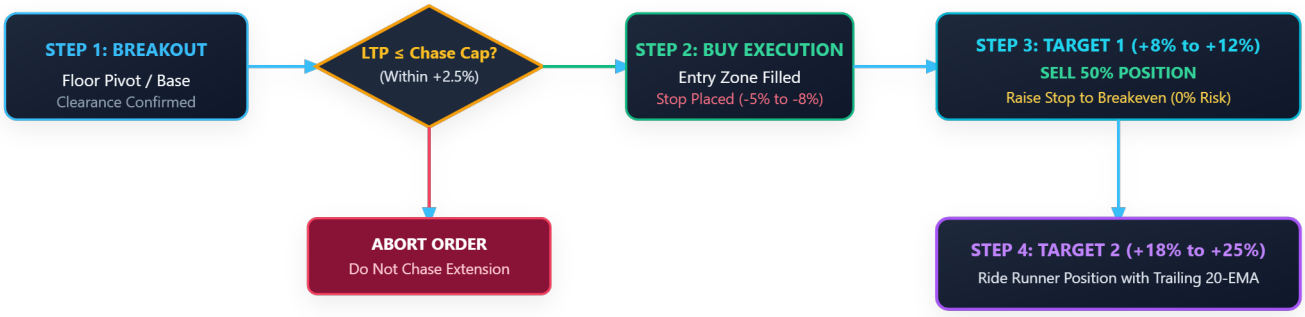
$$\text{OBIR} = \frac{\text{Total Bid Quantity (Levels 1–5)}}{\text{Total Ask Quantity (Levels 1–5)}}$$

- **Interpretation**:
- $OBIR > 2.0x$: Strong immediate buy-side liquidity cushion.
- $OBIR < 0.5x$: Overhead sell resistance dominating the book.

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4. Entry / Exit Radar, Execution Architecture & Risk Limits

FIGURE 4.1 — ENTRY / EXIT RADAR EXECUTION ARCHITECTURE



4.1 Breakout Pivot & Chase Cap

- **Breakout Pivot**: The exact horizontal price level that must be cleared on above-average volume ($RVOL \geq 1.4x$) to confirm an entry.
- **Chase Cap**: Strictly set at $Pivot + 2.5\%$.
- If a pivot is Rs. 400, the Chase Cap is Rs. 410.
- If price opens at Rs. 418 (+4.5%), entry is **gated**. Do not buy extended stocks.

4.2 Invalidation Floor (Hard Stop-Loss)

- **Definition**: The predetermined price level where the trade setup is mathematically broken (typically placed below the prior swing low or support shelf, -5% to -8%).
- **Mandatory Rule**: If LTP touches your hard stop, exit unconditionally. Never cancel or widen a stop-loss.

4.3 Multi-Horizon Targets & Breakeven Defense

- **Target 1 (T1: $+8|\%$ to $+12|\%$)**:
- **Mandatory Action**: Sell **50% of your shares** to bank cash profit.
- Immediately raise the stop-loss on the remaining 50% shares to your **exact purchase entry price (Breakeven)**.
- *Result*: The trade can never turn into a financial loss.
- **Target 2 (T2: $+18|\%$ to $+25|\%$)**: Ride the runner position to major Fibonacci extension levels.

4.4 Risk-to-Reward Ratio (RRR)

- **Definition**:

$$\text{RRR} = \frac{\text{Target 1 Price - Entry Price}}{\text{Entry Price - Stop-Loss Price}}$$

- **Rule**: Minimum permissible RRR is $1:2.0$ (ideally $1:2.5$ or $1:3.0$). Never enter trades offering less than Rs. 2 of reward per Rs. 1 of risk.

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5. Floor Pivots & Fibonacci Geometric Analysis

5.1 Classic Floor Pivots (P, R_1–R_3, S_1–S_3)

Calculated from the previous session's High (H), Low (L), and Close (C):

$$\text{Central Pivot (P)} = \frac{\text{H} + \text{L} + \text{C}}{3}$$

Level	Formula	Trading Function
R_3	$H + 2(P - L)$	Extreme intraday overbought extension.
R_2	$P + (H - L)$	Secondary profit target / heavy institutional supply.
R_1	$(2 \times P) - L$	First upward resistance barrier.
Pivot (P)	$(H + L + C) / 3$	Intraday equilibrium. Above = Bullish; Below = Bearish.
S_1	$(2 \times P) - H$	First downward support shelf.
S_2	$P - (H - L)$	Secondary institutional buying zone.
S_3	$L - 2(H - P)$	Extreme oversold intraday liquidity floor.

5.2 Fibonacci Retracement & Extension Geometry

Fibonacci Level	Technical Designation	Market Behavior & Execution Rule
$23.6|\%$	Shallow Retracement	Observed only in runaway, aggressive institutional trends.

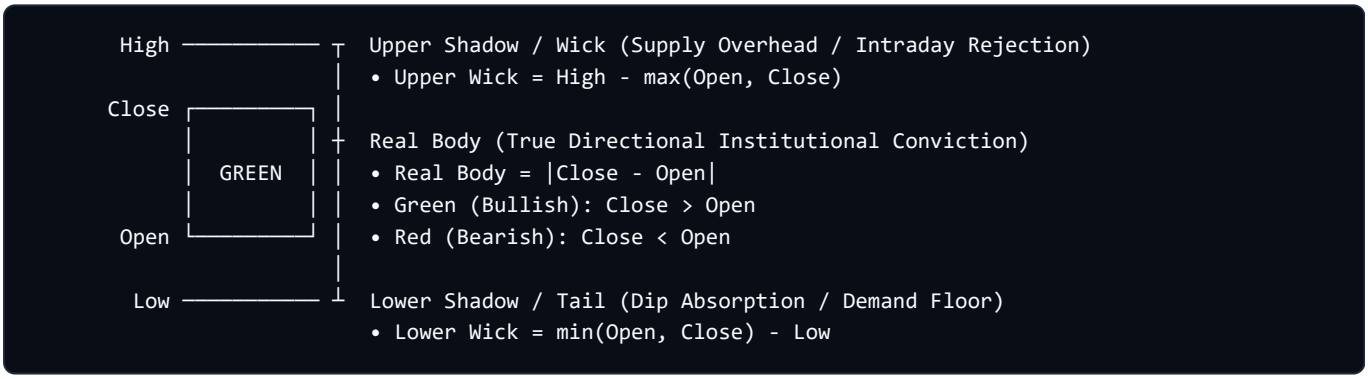
Fibonacci Level	Technical Designation	Market Behavior & Execution Rule
38.2%	Moderate Retracement	Standard healthy first pullback in a primary bull market.
50.0%	Equilibrium (Halfway)	Crucial Dow Theory psychological balance point.
61.8%	**Golden Ratio (Φ)**	**Highest probability institutional reversal zone.** Primary entry zone upon reversal candlestick confirmation.
78.6%	Deep Retracement	Final defense line before base failure.
161.8%	**Golden Extension**	Target projection for stocks breaking out into all-time highs.

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6. Candlestick Anatomy & The 28 Japanese Candlestick Patterns

6.1 Anatomical Structure & The Physics of a Candlestick

Every Japanese candlestick contains 4 fundamental price points (OHLC) that convey the battle between buying demand and selling supply:



The 5 Primary Candlestick Metrics Calculated in Code:

1. **Total Candle Range (Spread)**:

$$\text{Range} = \text{High} - \text{Low}$$

2. **Real Body Ratio (R_{body})**:

$$R_{body} = \frac{|\text{Close} - \text{Open}|}{\text{High} - \text{Low}}$$

3. $R_{body} \geq 0.82$: Marubozu (one-way institutional momentum).
4. $0.25 \leq R_{body} \leq 0.55$: Standard balanced trading.
5. $R_{body} \leq 0.14$: Doji (equilibrium, supply/demand standoff).
6. **Upper Wick Ratio (R_{upper})**:

$$R_{upper} = \frac{\text{High} - \max(\text{Open}, \text{Close})}{\text{High} - \text{Low}}$$

7. $R_{upper} \geq 0.62$: Massive overhead selling rejection (Shooting Star / Gravestone Doji).
8. **Lower Tail Ratio (R_{lower})**:

$$R_{lower} = \frac{\min(Open, Close) - Low}{High - Low}$$

9. $R_{lower} \geq 0.62$: Strong institutional buying absorption (Hammer / Dragonfly Doji).
10. **Gap Distance ($|\Delta_{gap}$)**:

$$\Delta_{gap} = \frac{Open - Close_{prev}}{Close_{prev}} \times 100\%$$

11. $|\Delta_{gap} \geq +1.5|\%$: Rising Window or Bullish Kicker impulse.
12. $|\Delta_{gap} \leq -1.5|\%$: Falling Window or Bearish Kicker shock.

6.2 The App's 3 Candlestick Recognition Engines

Drabyashree NEPSE utilizes 3 complementary, deterministic recognition engines (`candlestickPatterns.js` and `priceActionEngine.js`): 1. **Live Feed Single-Candle Scanner (`detectSingleStockPatterns`)**: Evaluates real-time NEPSE live trading ticks instantly against geometric body/wick ratios. 2. **Historical Chronological Scanner (`detectCandlestickPatternsFromHistory`)**: Iterates through the stock's last 45 daily sessions, evaluating 3-candle, 2-candle, and 1-candle sequences chronologically (highest specificity first). 3. **Price Action Structural Engine (`analyzeCandlestickPattern`)**: Evaluates candle shapes in direct relation to swing highs/lows, support/resistance shelves, and relative volume (RVOL).

6.3 Complete Master Catalog of All 28 Japanese Candlestick Patterns

Group A: Bullish Reversal Patterns (Bottom Accumulation & Rebound)

#	Pattern Name & Nepali Name	ID & Candles	Mathematical Detection Algorithm in Code	Market Psychology & Who is Winning	Actionable Rule & Invalidation Level	Reliability
1	**Bullish Engulfing** *(बुलिश एनाल्फिङ)*	`bullish_engulfing` (2 Candles)	`prevIsBear && isBull && open <= prevClose * 0.995 && close >= prevOpen * 1.018 && body > prevBody * 1.05`	Bears controlled session 1, but session 2 opens lower and unleashes massive institutional buying that swallows previous day's supply.	Buy above green candle high. Place stop-loss 1 tick below green candle low.	⭐⭐⭐ (High)

#	Pattern Name & Nepali Name	ID & Candles	Mathematical Detection Algorithm in Code	Market Psychology & Who is Winning	Actionable Rule & Invalidation Level	Reliability
2	**Hammer** *(ह्यामर - तल्लो रिभर्सल)*	`hammer` (1 Candle)	`lowerWick >= body * 1.9 && upperWick <= body * 0.55 && (pChange >= -0.8 || isBull)`	Intraday panic dumping pushed prices to session lows, where whales stepped in and absorbed all floating supply before close.	Buy on next-day follow-through above Hammer high. Stop-loss anchored below Hammer tail.	⭐⭐⭐ (High)
3	**Inverted Hammer** *(इन्भर्टेड ह्यामर)*	`inverted_hammer` (1 Candle)	`upperWick >= body * 1.9 && lowerWick <= body * 0.55 && isBull && pChange >= 0`	Formed after a downtrend. Bulls launched an aggressive upward reconnaissance test. Price settled off highs but held green.	Await next-day confirmation candle closing above Inverted Hammer body before entering.	⭐⭐ (Medium)
4	**Morning Star** *(मर्निङ स्टार - ३-क्यान्डल)*	`morning_star` (3 Candles)	`!p2Bull && p2Body > p1Body * 1.8 && isBull && close >= (p2.open + p2.close) / 2`	Large red candle (bear dominance) |*rightarrow* small star candle (selling exhaustion) |*rightarrow* strong green candle (buyers regain dominance).	High-conviction swing entry. Stop-loss placed strictly below the middle star candle low.	⭐⭐⭐ (High)
5	**Piercing Pattern** *(पियर्सिङ प्याटर्न)*	`piercing_pattern` (2 Candles)	`!p1Bull && isBull && open < p1.low && close > (p1.open + p1.close) / 2 && close < p1.open`	Day 2 gaps down below Day 1 low, triggering retail stop-losses. Institutions buy the panic, driving price above 50% of Day 1 body.	Buy on confirmation. Stop-loss placed 1 tick below the two-candle low.	⭐⭐⭐ (High)
6	**Tweezer Bottom** *(ट्विजर्स बटम - डबल सपोर्ट)*	`tweezer_bottom` (2 Candles)	`!p1Bull && isBull && Math.abs(low - p1.low) / low <= 0.004 && pChange > 0.5`	Two consecutive sessions test the exact same price floor and bounce. Confirms institutional bid wall defense.	Enter immediately with tight stop-loss 1.5% below the dual-session support floor.	⭐⭐⭐ (High)

#	Pattern Name & Nepali Name	ID & Candles	Mathematical Detection Algorithm in Code	Market Psychology & Who is Winning	Actionable Rule & Invalidation Level	Reliability
7	**Three White Soldiers** *(थ्री ह्वाइट सोल्जर्स)*	`three_white_soldiers` (3 Candles)	`p2Bull && p1Bull && isBull && close > p1.close && p1.close > p2.close && bodyRatio >= 0.55`	Relentless, sustained institutional buying across 3 sessions with progressive higher opens and closes.	Ride the primary trend. Do not chase if extended $>3x$ ATR; buy first shallow 1-2 day pullback to 20-EMA.	⭐⭐⭐ (High)
8	**Bullish Harami** *(बुलिश हरामी - इनसाइड बार)*	`bullish_harami` (2 Candles)	`!p1Bull && isBull && high <= p1.open && low >= p1.close && body < p1Body * 0.6`	Small green body sits entirely inside prior large red body. Downside momentum abruptly halts into tight equilibrium.	Enter on breakout above mother candle high. Place stop-loss beneath the inside bar low.	⭐⭐ (Medium)
9	**Dragonfly Doji** *(ड्रयागनफ्लाई डोजी)*	`dragonfly_doji` (1 Candle)	`bodyRatio <= 0.14 && lowerRatio >= 0.62 && upperRatio <= 0.18`	Open, High, and Close converge at the top of the candle with a deep lower shadow. Bears completely failed to hold lows.	Bullish reversal anchor when formed at support or 50-EMA. Stop-loss below lower shadow.	⭐⭐⭐ (High)
10	**Abandoned Baby Bull** *(एबान्डन्ड बेबी)*	`abandoned_baby` (3 Candles)	`!p2Bull && p1Body < p2Body * 0.3 && isBull && p1.high < p2.low && p1.high < close.low`	Rare island reversal: Red candle $\|rightarrow$ gapped down isolated Doji $\|rightarrow$ sharp gapped-up green candle.	Immediate high-probability swing entry. Place stop-loss just beneath the isolated Doji low.	⭐⭐⭐ (High)
11	**Bullish Kicker** *(बुलिश किकर)*	`bullish_kickers` (2 Candles)	`!p1Bull && isBull && open >= p1.open && pChange >= 3.0 && bodyRatio >= 0.6`	Dramatic positive catalyst. Price gaps open above prior red candle open and drives upward with heavy volume.	Aggressive trend continuation entry. Buy breakout or first intraday dip toward gap level.	⭐⭐⭐ (High)
12	**Rising Window (Gap Up)** *(राइजिङ विन्डो)*	`rising_window` (2 Candles)	`low > p1.high * 1.003 && isBull && pChange >= 2.2`	Void between prior day high and current low. Overwhelming buying demand outstripping circulating float.	Gap window functions as strong magnetic support on any future retracements.	⭐⭐⭐ (High)

Group B: Bearish Reversal Patterns (Top Distribution & Exhaustion)

#	Pattern Name & Nepali Name	ID & Candles	Mathematical Detection Algorithm in Code	Market Psychology & Who is Winning	Actionable Rule & Invalidation Level	Reliability
13	**Bearish Engulfing** *(बियरिश एनाल्फिङ)*	`bearish_engulfing` (2 Candles)	`p1Bull && !isBull && open >= p1.close && close <= p1.open && body > p1Body * 1.05`	Session opens above prior green close, but aggressive institutional selling overwhelms all retail bids, closing below prior open.	Exit swing trades or tighten stops to breakeven. Major top reversal warning.	⭐⭐⭐ (High)
14	**Shooting Star** *(शुटिङ स्टार - माथिल्लो रिजेक्सन)*	`shooting_star` (1 Candle)	`upperWick >= body * 1.9 && lowerWick <= body * 0.55 && (pChange < 0 || !isBull)`	Price surged intraday to test resistance, but whales dumped heavy supply, knocking price back down to session lows.	Strict profit-taking alert. If holding, place stop-loss at low of Shooting Star bar.	⭐⭐⭐ (High)
15	**Hanging Man** *(ह्याङ्गिङ म्यान)*	`hanging_man` (1 Candle)	`lowerWick >= body * 1.9 && upperWick <= body * 0.55 && !isBull && p1 && close < p1.close`	Appears at top of an uptrend. Long lower tail proves sellers are breaking through the trend's foundation.	Warning of trend rollover. Avoid fresh buying; trim exposure if next candle breaks low.	⭐⭐ (Medium)
16	**Evening Star** *(इभिनिङ स्टार)*	`evening_star` (3 Candles)	`p2Bull && p2Body > p1Body * 1.8 && !isBull && close <= (p2.open + p2.close) / 2`	Large green candle $\rightarrow$ exhausted star candle at high $\rightarrow$ deep red breakdown candle penetrating prior advance.	Classic institutional distribution top. Sell immediately or hedge long exposure.	⭐⭐⭐ (High)
17	**Dark Cloud Cover** *(डार्क क्लाउड कभर)*	`dark_cloud_cover` (2 Candles)	`p1Bull && !isBull && open > p1.high && close < (p1.open + p1.close) / 2 && close > p1.open`	Whales sell into an opening gap up, forcing price deep into previous day's green body (closing below 50% midpoint).	Trim swing positions. Stiff overhead resistance confirmed.	⭐⭐⭐ (High)

#	Pattern Name & Nepali Name	ID & Candles	Mathematical Detection Algorithm in Code	Market Psychology & Who is Winning	Actionable Rule & Invalidation Level	Reliability
18	**Tweezer Top** *(ट्विजर्स टप)*	`tweezer_top` (2 Candles)	`p1Bull && !isBull && Math.abs(high - p1.high) / high <= 0.004 && pChange < -0.5`	Two consecutive sessions rejected at the exact same ceiling. Whales defending an institutional sell wall.	Take profits at resistance unless price breaks ceiling on $>2x$ volume.	⭐⭐⭐ (High)
19	**Three Black Crows** *(थ्री ब्ल्याक क्रोज)*	`three_black_crows` (3 Candles)	`!p2Bull && !p1Bull && !isBull && close < p1.close && p1.close < p2.close && bodyRatio >= 0.55`	Severe, systematic institutional dumping across 3 sessions closing near absolute daily lows.	Do not catch falling knives. Stay in cash until base formation completes.	⭐⭐⭐ (High)
20	**Bearish Harami** *(बियरिश हरामी)*	`bearish_harami` (2 Candles)	`p1Bull && !isBull && high <= p1.close && low >= p1.open && body < p1Body * 0.6`	Small red candle trapped inside prior large green body. Buying volume dried up completely at the peak.	Tighten trailing stops to mother candle low; prepare for downward break.	⭐⭐ (Medium)
21	**Gravestone Doji** *(ग्रेभस्टोन डोजी)*	`gravestone_doji` (1 Candle)	`bodyRatio <= 0.14 && upperRatio >= 0.62 && lowerRatio <= 0.18`	Open, Low, and Close sit at bottom of range with massive upper shadow. Bulls lost 100% of intraday gains.	Extreme overhead rejection warning. Liquidate speculative momentum positions.	⭐⭐⭐ (High)
22	**Abandoned Baby Bear** *(एबान्डन्ड बेबी - टप)*	`abandoned_baby_bear` (3 Candles)	`p2Bull && p1Body < p2Body * 0.3 && !isBull && p1.low > p2.high && p1.low > close.high`	Rare island top: Green advance $\rightarrow$ gapped-up Doji $\rightarrow$ gapped-down red candle leaving Doji trapped.	High-risk bull trap. Exit all swing positions unconditionally.	⭐⭐⭐ (High)

#	Pattern Name & Nepali Name	ID & Candles	Mathematical Detection Algorithm in Code	Market Psychology & Who is Winning	Actionable Rule & Invalidation Level	Reliability
23	**Bearish Kicker** *(बियरिश किकर)*	`bearish_kickers` (2 Candles)	`p1Bull && !isBull && open <= p1.open && pChange <= -3.0 && bodyRatio >= 0.6`	Sudden negative shock. Price gaps down below prior green open and sells off with high volume.	Emergency exit signal. Respect the gap-down resistance zone.	⭐⭐⭐ (High)
24	**Falling Window (Gap Down)** *(फलिङ विन्डो)*	`falling_window` (2 Candles)	`high < p1.low * 0.997 && !isBull && pChange <= -2.2`	Void between prior day low and current high. Aggressive supply overwhelm and liquidity vacuum.	Falling window gap becomes impenetrable overhead resistance on any bounce.	⭐⭐⭐ (High)

Group C: Institutional Momentum Expansion Patterns

#	Pattern Name & Nepali Name	ID & Candles	Mathematical Detection Algorithm in Code	Market Psychology & Who is Winning	Actionable Rule & Invalidation Level	Reliability
25	**Bullish Marubozu** *(बुलिश मारुबोजु)*	`bullish_marubozu` (1 Candle)	`bodyRatio >= 0.82 && pChange >= 2.0 && isBull`	Complete, relentless buyer control from the opening bell to the close with negligible wicks ($\leq 18\%$).	Strong breakout confirmation. Candle midpoint (50%) becomes prime support on dips.	⭐⭐⭐ (High)
26	**Bearish Marubozu** *(बियरिश मारुबोजु)*	`bearish_marubozu` (1 Candle)	`bodyRatio >= 0.82 && pChange <= -2.0 && !isBull`	Complete panic selling and institutional liquidation from open to close without buyer pushback.	Never buy after a Bearish Marubozu. Expect further downward drift.	⭐⭐⭐ (High)

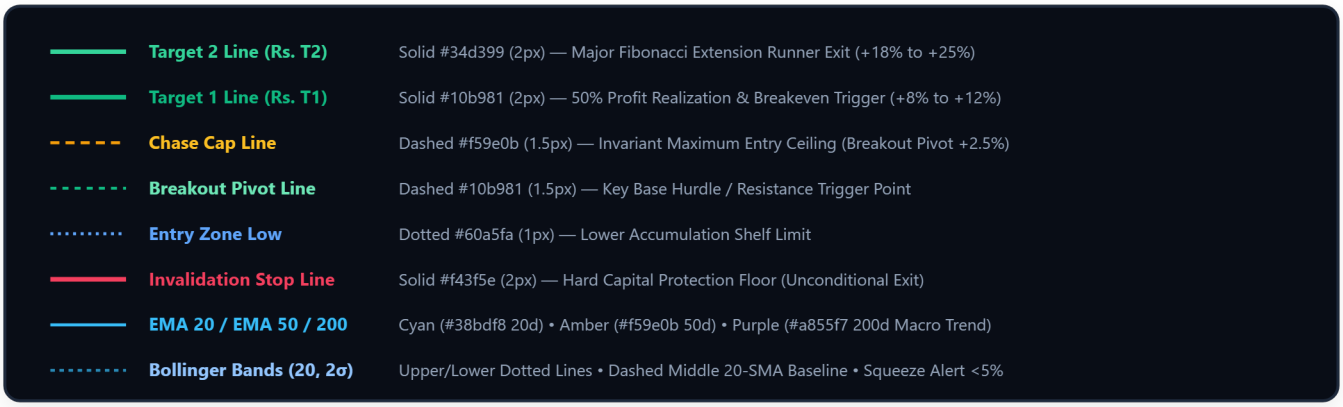
Group D: Indecision, Equilibrium & Consolidation Patterns

#	Pattern Name & Nepali Name	ID & Candles	Mathematical Detection Algorithm in Code	Market Psychology & Who is Winning	Actionable Rule & Invalidation Level	Reliability
27	**Classic Doji** *(क्लासिक डोजी)*	`doji` (1 Candle)	`bodyRatio <= 0.14 && (range / close) >= 0.01 && upperRatio < 0.62 && lowerRatio < 0.62`	Open and Close are virtually identical. Perfect equilibrium and standoff between bulls and bears.	Neutral alone. Await next candle breakout above Doji high (buy) or below Doji low (sell).	⭐⭐ (Medium)
28	**Long-Legged Doji & Spinning Top** *(लङ्-लेग्ड डोजी र स्पिनिङ् टप्)*	`long_legged_doji` / `spinning_top` (1 Candle)	`(bodyRatio <= 0.14 && range / close >= 0.035)` or `(bodyRatio between 0.14 and 0.34 && upperRatio >= 0.26 && lowerRatio >= 0.26)`	High volatility with rapid swings in both directions, but market closes with no net directional progress.	Signals imminent volatility expansion. Enter in direction of next session's volume breakout.	⭐⭐ (Medium)

7. Interactive Chart Lines, Overlays & Technical Geometry

In Drabyashree NEPSE, chart screens (`ShareHubChart.jsx` , `StockCandlestickChart.tsx` , and `AdvancedChartModal.jsx`) render technical geometry using high-performance Canvas and SVG pipelines. Every line has a strict quantitative meaning:

FIGURE 7.1 — INTERACTIVE CHART OVERLAY LINES & GEOMETRY



7.1 Complete Guide to Chart Lines & Colors

Visual Element	Color Code & Style	Technical Formula	Quantitative Purpose & Investor Action
Candlestick (Bull)	`#10b981` (Emerald Green)	$Close \geq Open$	Session closed higher than it opened. Buying pressure dominated.

Visual Element	Color Code & Style	Technical Formula	Quantitative Purpose & Investor Action
Candlestick (Bear)	`#f43f5e` (Rose Red)	$Close < Open$	Session closed lower than it opened. Selling pressure dominated.
EMA 20 Line	`#38bdf8` (Cyan), 1.5px Solid	$EMA_{20} = (Close - EMA_{t-1}) \times \frac{2}{21} + EMA_{t-1}$	**Short-Term Momentum Guide**. In strong trends, price remains above EMA-20. Pullbacks touching EMA-20 are prime add-on opportunities.
EMA 50 Line	`#f59e0b` (Amber), 1.5px Solid	$EMA_{50} = (Close - EMA_{t-1}) \times \frac{2}{51} + EMA_{t-1}$	**Institutional Defense Line**. Mutual funds systematically accumulate around the rising 50-EMA. If price closes below 50-EMA for 2+ days, exit swing trades.
EMA 200 Line	`#a855f7` (Purple), 2.0px Solid	$EMA_{200} = (Close - EMA_{t-1}) \times \frac{2}{201} + EMA_{t-1}$	**Macro Regime Filter**. Stocks above EMA-200 are in structural bull markets. Stocks below EMA-200 are in bear cycles. Never hold swing positions below EMA-200.
Bollinger Upper Band	`rgba(56, 189, 248, 0.65)` Dotted	$SMA_{20} + (2 \times \sigma)$	Extreme statistical overbought barrier ($+2\sigma$). When tagged during low volume, expect pullback to middle band.
Bollinger Middle Band	`rgba(148, 163, 184, 0.5)` Dashed	$SMA_{20} = \frac{1}{20} \sum_{i=1}^{20} C_i$	Dynamic trend equilibrium. Acts as support in bull trends and resistance in bear trends.
Bollinger Lower Band	`rgba(56, 189, 248, 0.65)` Dotted	$SMA_{20} - (2 \times \sigma)$	Extreme statistical oversold barrier (-2σ). Confluence with Hammer indicates institutional buying absorption.
Support PriceLine	`rgba(52, 211, 153, 0.75)` Dashed	Confluence Cluster ($< LTP$)	Key demand floor with multiple historical touches. Place buy limits slightly above this line.
Resistance PriceLine	`rgba(251, 113, 133, 0.75)` Dashed	Confluence Cluster ($> LTP$)	Heavy overhead supply shelf. Book profits or await high-volume breakout clearance.
Breakout Pivot Line	`#10b981` Dashed	Prior Base Ceiling	The trigger line. A decisive close above this line activates a trade entry.
Chase Cap Line	`#f59e0b` Dashed	$Breakout\ Pivot \times 1.025$	**Strict execution ceiling**. Never buy above this line ($+2.5\%$ limit). Protects against mean-reversion drawdowns.
Invalidation Stop Line	`#f43f5e` Solid 2px	Prior Swing Low / Support Shelf	**Hard capital preservation line**. If touched, exit 100% unconditionally.
Target 1 Line (T1)	`#10b981` Solid 2px	Entry $+8\%$ to $+12\%$	First profit milestone. Sell 50% shares and raise remaining stop to entry price.
Target 2 Line (T2)	`#34d399` Solid 2px	Entry $+18\%$ to $+25\%$	Secondary runner milestone. Ride until trailing 20-EMA breaks.
Volume Up Bar	`rgba(16, 185, 129, 0.25)`	$Close \geq Open$	Institutional accumulation bar. Height indicates trade volume.
Volume Down Bar	`rgba(244, 63, 94, 0.25)`	$Close < Open$	Liquidation bar. If tall ($>2x$ average), indicates distribution.

7.2 Intraday 1-Minute to 5-Minute Aggregation Engine

When selecting the **1D Timeframe** in `ShareHubChart.jsx`: * The app automatically aligns to **Nepal Standard Time (NPT: UTC+5:45, offset +20,700s)**. * Aggregates raw 1-minute exchange ticks into authentic **5-minute OHLC candlesticks** across the official 11:00 AM to 3:00 PM NEPSE trading session. * Pinned volume scale occupies the lower 20% of the canvas (`top: 0.8, bottom: 0`) to prevent volume spikes from obscuring price action.

7.3 Interactive Crosshair & Inspection Tooltip Readout

Hovering or tapping anywhere on the chart activates the dashed crosshair coordinates and displays an instantaneous technical audit: * **Date**: ISO formatted Nepal trading date. * **Open, High, Low, Close (OHLC)**: Exact Rupee prices. * **Change %**: Intraday price movement ($\frac{Close - Open}{Open} \times 100\%$). * **Volume**: Total kitta traded in the active period. * **Matching EMA-20 & EMA-50**: Precise moving average values at the hovered timestamp, allowing instant verification of whether price is above or below institutional trendlines.

7.4 Market Structure Swings & Breakout Validation Geometry

In `src/utils/priceActionEngine.js`, market structure is classified using fractal swing identification:

1. **Fractal Swing Highs & Lows**:
2. Evaluated across a 2-bar left and right lookback window.
3. `Swing High`: Bar high is strictly higher than 2 bars before and 2 bars after.
4. `Swing Low`: Bar low is strictly lower than 2 bars before and 2 bars after.
5. **The 5 Structural Regimes**:
6. **Higher Highs & Higher Lows (Strength: 82)**: Confirmed Bullish Market Structure.
7. **Lower Highs & Lower Lows (Strength: 20)**: Confirmed Bearish Downtrend.
8. **Tight Consolidation Range (Strength: 60)**: Price oscillating inside a $\leq 4.2\%$ band across 15 sessions. Precursor to explosive breakout.
9. **Contracting Triangle Compression (Strength: 65)**: Converging lower highs and higher lows. Volatility coiling.
10. **Broadening Expansion (Strength: 45)**: Expanding highs and expanding lows. Unstable two-way distribution.
11. **Confluence Support & Resistance Clustering**:
12. Gathers candidates from 5 sources: Swing pivots, 52-week High/Low, 20-day range extremes, Classic Floor Pivots (P, R_1, S_1, R_2, S_2), and Fibonacci Retracements ($38.2\%, 50\%, 61.8\%$).
13. Clusters candidates within a strict 1.2% **price tolerance**. Multi-source clusters receive the highest strength scores (45 to 95).
14. **Breakout Clearance Buffer & Bull Trap Detection**:
15. **Clearance Buffer**: To prevent 10-paisa noise from triggering false breakouts, price must clear resistance by:

$$\text{Clearance Buffer} = \max(\text{LTP} \times 0.35\%,\ 0.20 \times \text{ATR})$$

16. **Bull Trap (Exhaustion Breakout)**: If a candle breaches resistance but its **upper wick constitutes $\geq 40\%$ of its total range**, whales sold into retail breakout buyers. High risk of immediate reversal.

17. **Breakout Retest**: Price breaks resistance, pulls back within $1.5|\%$ of the broken level, and bounces with green follow-through $|rightarrow$ Highest-conviction entry point (Strength: 88).

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8. Volatility, Statistical Momentum & Technical Speedometer

8.1 The 12-Indicator Technical Rating Speedometer Engine

Located in `StockTechnicalGaugeCard.jsx` and `candlestickPatterns.js`, the app synthesizes **6 Moving Averages + 6 Oscillators** into an institutional speedometer needle ($-88^|circ$ to $+88^|circ$) and consensus rating:

FIGURE 8.1 — TECHNICAL SPEEDOMETER CONSENSUS ARCHITECTURE



Speedometer Signal Evaluation Matrix:

Category	Indicator	Buy Signal Trigger	Neutral Zone	Sell Signal Trigger
Moving Average	**EMA 10**	$LTP > EMA_{10} + 0.35|\%$	Within $|pm 0.35|\%$	$LTP < EMA_{10} - 0.35|\%$
Moving Average	**EMA 20**	$LTP > EMA_{20} + 0.35|\%$	Within $|pm 0.35|\%$	$LTP < EMA_{20} - 0.35|\%$
Moving Average	**SMA 50**	$LTP > SMA_{50} + 0.35|\%$	Within $|pm 0.35|\%$	$LTP < SMA_{50} - 0.35|\%$
Moving Average	**EMA 50**	$LTP > EMA_{50} + 0.35|\%$	Within $|pm 0.35|\%$	$LTP < EMA_{50} - 0.35|\%$
Moving Average	**SMA 100**	$LTP > SMA_{100} + 0.35|\%$	Within $|pm 0.35|\%$	$LTP < SMA_{100} - 0.35|\%$
Moving Average	**SMA 200**	$LTP > SMA_{200} + 0.35|\%$	Within $|pm 0.35|\%$	$LTP < SMA_{200} - 0.35|\%$
Oscillator	**RSI (14)**	$52 \le RSI \le 72$ or $RSI < 32$	$44 \le RSI < 52$	$RSI > 75$ or $RSI < 44$
Oscillator	**MACD Histogram**	$Histogram > +0.05$	$-0.05 \le Histo \le +0.05$	$Histogram < -0.05$
Oscillator	**Stochastic %K**	$50|\% \le |\%K \le 82|\%$ or $|\%K < 22|\%$	$40|\% \le |\%K < 50|\%$	$|\%K > 88|\%$ or $|\%K < 40|\%$

Category	Indicator	Buy Signal Trigger	Neutral Zone	Sell Signal Trigger
Oscillator	**Volume RVOL (20D)**	$RVOL \geq 1.25x$ with $\|Delta\ P \geq 0$	$0.80x \leq RVOL < 1.25x$	$RVOL \geq 1.25x$ with $\|Delta\ P < -1\|\%$
Oscillator	**ADX Trend**	$ADX \geq 22$ with $\|Delta\ P > 0$	$ADX < 22$ (No Trend)	$ADX \geq 22$ with $\|Delta\ P < 0$
Oscillator	**CCI (20)**	$CCI > +25$	$-25 \leq CCI \leq +25$	$CCI < -25$

Consensus Rating Output:

$$\text{Consensus Ratio} = \frac{\text{Total Buys - Total Sells}}{\text{Total Evaluated Indicators}} \quad (-1.0 \text{ to } +1.0)$$

- **Strong Buy**: $Consensus\ Ratio \geq +0.50$ (Needle angle $+44^\circ$ to $+88^\circ$).
- **Buy**: $+0.16 \leq Consensus\ Ratio < +0.50$ (Needle angle $+14^\circ$ to $+44^\circ$).
- **Neutral**: $-0.16 < Consensus\ Ratio < +0.16$ (Needle centered around 0°).
- **Sell**: $-0.50 < Consensus\ Ratio \leq -0.16$ (Needle angle -14° to -44°).
- **Strong Sell**: $Consensus\ Ratio \leq -0.50$ (Needle angle -44° to -88°).

8.2 Bollinger Bands ($20,\ 2\sigma$) & Bollinger Squeeze

- **Middle Band**: 20-period Simple Moving Average (SMA_{20}).
- **Upper / Lower Bands**: $Middle\ Band \pm (2 \times Standard\ Deviation)$.
- **Bandwidth ($\%BW$)**:

$$\%BW = \frac{\text{Upper Band - Lower Band}}{\text{Middle Band}} \times 100\%$$

- **Bollinger Squeeze**: Occurs when $\%BW < 5.0\%$. Signals extreme volatility compression; an explosive $+15\%$ to $+30\%$ move is imminent.
- **$\%B$ Indicator**:

$$\%B = \frac{\text{LTP - Lower Band}}{\text{Upper Band - Lower Band}} \times 100\%$$

- $\%B > 100\%$: Price closed above upper band (strong momentum, but elevated pullback risk).
- $\%B < 0\%$: Price closed below lower band (panic liquidation).

8.3 ATR (Average True Range — 14 Periods)

- **Definition**: Measures absolute price volatility in Nepalese Rupees using Wilder's smoothed method:

$$TR = \max\big((H - L),\ |H - C_{prev}|,\ |L - C_{prev}|\big)$$

- **Usage**: Set dynamic stop-losses at $1.5x$ to $2.0x$ ATR below purchase price to prevent being shaken out by normal daily market noise.

8.4 Volume Z-Score (Z_{vol})

- **Definition**: Measures how many standard deviations today's volume deviates from its 20-day rolling mean:

$$Z_{vol} = \frac{Volume - \mu_{20}}{\sigma_{20}}$$

- **Interpretation**:
- $Z_{vol} > +2.0$: Statistically significant institutional volume footprint ($97.7th$ percentile).
- $Z_{vol} < -1.0$: Extreme volume dry-up (VDU).

8.5 Mansfield Relative Strength (MRS)

- **Definition**: Measures a stock's outperformance relative to the benchmark NEPSE Index:
- $MRS > 0$: **Market Leader**. Outperforming NEPSE; leads the market upward during bull phases.
- $MRS < 0$: **Laggard**. Underperforming the index; avoid deploying swing capital here.

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9. Fundamentals, Solvency & Valuation Frameworks

9.1 EPS, P/E & PBV Multiples

- **EPS (Earnings Per Share)**: $Net\ Profit\ After\ Tax\ /\ Total\ Outstanding\ Shares$.
- **P/E (Price-to-Earnings)**: $LTP\ /\ EPS$. Compare against Sector P/E.
- **PBV (Price-to-Book Value)**: $LTP\ /\ Book\ Value\ Per\ Share\ (BVPS)$. Measures asset backing.

9.2 Banking Solvency: NPL & CAR (NRB Regulatory Directives)

- **NPL (Non-Performing Loan Ratio)**: Percentage of loans overdue > 90 days.
- **NPL** $\leq 3.5|\%$: Healthy bank.
- **NPL** $> 5.0|\%$: **Severe Red Flag.** Nepal Rastra Bank **prohibits banks with** $NPL > 5.0|\%$ **from distributing any cash dividends or bonus shares**.
- **CAR (Capital Adequacy Ratio — Basel III)**:
- Must be maintained at $\geq 11.5|\%$.
- If CAR drops below $11.5|\%$, NRB imposes Prompt Corrective Action (PCA) and halts all dividend declarations.

9.3 Graham Intrinsic Valuation & Margin of Safety

- **Classical Formula**:

$$\text{Intrinsic Value} = \sqrt{22.5 \times \text{EPS} \times \text{BVPS}}$$

$$\text{Margin of Safety (MoS)} = \frac{\text{Intrinsic Value - LTP}}{\text{Intrinsic Value}} \times 100\%$$

- **Thresholds**:
- $MoS \geq +20\%$: Deeply undervalued with tangible asset protection.
- $MoS < -20\%$: Premium multiple; trade purely on momentum, not value.

9.4 Equity Risk Premium (ERP vs. Bank Fixed Deposit)

- **Definition**:

$$\text{Earnings Yield} = \frac{\text{EPS}}{\text{LTP}} \times 100\% \quad | \quad \text{ERP} = \text{Earnings Yield - Bank FD Rate (7.5\%)}$$

- **Warning**: If a stock has an earnings yield of 3.5% and commercial banks pay 7.5% on Fixed Deposits, $ERP = -4.0\%$. You are accepting stock market volatility for less return than a risk-free bank deposit.

9.5 Buffett Indicator (NEPSE Capitalization to GDP)

- **Formula**: $\frac{\text{Total NEPSE Market Cap}}{\text{Nepal GDP}} \times 100\%$.
- $< 60\%$: Macro undervaluation (generational buying zone).
- $70\% - 100\%$: Fair macro valuation.
- $> 120\%$: Macro bubble territory (seen at NEPSE 3,200 in 2021). Expect severe systemic correction.

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10. Capital Structure, Public Float & Corporate Actions

10.1 Public Float vs. Float Market Cap

- **Public Float**: The actual percentage of shares tradable on the exchange floor (typically 30% to 49%, with promoters holding 51% to 70%).
- **Float Market Cap**: $\text{Public Float Shares} \times LTP$.
- **Market Dynamics**: Low float stocks ($< \text{Rs. 50 Crores}$) can be easily cornered by syndicates, causing rapid price movements, but carry severe liquidity exit risk.

10.2 Promoter Lock-in Expiry (The 3-Year Liquidity Shock)

- **Statutory Rule**: SEBON mandates that promoter shares of listed companies (especially Hydropower and Manufacturing) are legally **locked for 3 years** following IPO allotment.
- **The Hazard**: When the 3-year lock-in date arrives, **millions of promoter shares unlock simultaneously**. Promoters who acquired shares at Rs. 100 often dump them onto the open market at Rs. 300–400, causing a severe supply shock and steep price collapse. Always check the **Promoter Lock-in Tool** before entering hydropower positions.

10.3 Bonus Share & Right Share Adjustment Formulas

- **Bonus Adjustment**:

$$\text{Adjusted Price} = \frac{\text{Previous Close}}{1 + \frac{\text{Bonus }\%}{100}}$$

- **Right Share Adjustment**:

$$\text{Adjusted Price} = \frac{\text{Previous Close} + (\text{Right Ratio} \times 100)}{1 + \text{Right Ratio}}$$

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11. Statutory Regulations, Taxes, Fees & Settlement Operations

11.1 T+2 Rolling Settlement Cycle

- **Mechanics**:
- **Day** T: Trade executed.
- **Day** $T+1$: Clearing & broker transfer.
- **Day** $T+2$: Shares credited to Demat; funds settled.
- **Strict Constraint**: You **cannot sell your shares on Day** T **or Day** $T+1$. If a stock drops after you buy, you are legally locked into holding the position for 2 business days.

11.2 Capital Gains Tax (CGT) Slabs

- **Individual Short-Term (** < 365 **days)**: $7.5|\%$ on net gain.
- **Individual Long-Term (** ≥ 365 **days)**: $5.0|\%$ on net gain.
- **Institutional / Corporate**: $10.0|\%$.

11.3 SEBON Brokerage Slabs & Statutory Costs

Every trade on NEPSE incurs statutory fees: * **SEBON Regulatory Fee**: $0.015|\%$ of turnover. * **DP (Depository Participant) Fee**: Flat **Rs. 25** per scrip per day (on sale). * **Brokerage Commission Rates**: * Up to Rs. 50,000: $0.40|\%$ * Rs. 50,000 to Rs. 500,000: $0.37|\%$ * Rs. 500,000 to Rs. 2,000,000: $0.34|\%$ * Rs. 2,000,000 to Rs. 1 Crore: $0.30|\%$ * Above Rs. 1 Crore: $0.27|\%$ * **Statutory Breakeven**: A stock must rise by at least $+0.80|\%$ **to** $+0.90|\%$ just to cover round-trip transaction costs.

11.4 NRB Margin Lending (70% LTV & Maintenance Rules)

- **Loan Ceiling**: Bank share loans are capped at $70|\%$ of the 180-day average price or current LTP (whichever is lower).
- **Individual Borrower Cap**: Rs. 15 Crores.
- **Margin Calls**: If the market drops and collateral coverage falls below 150%, banks issue a margin call. If not topped up within the deadline, the bank will force-sell your shares at market price.

11.5 15% Daily Circuit Band

- Under SEBON Fourth Amendment 2082, the maximum daily price fluctuation on individual equities is $|pm 15|\%$.

- An upper circuit lock ($+15\%$) means there are zero sellers; a lower circuit lock (-15%) means there are zero buyers.

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12. Calendar Cycles, Corporate Timing & Seasonality

12.1 Book Closure Date vs. Record Date

- **Rule**: You must buy and hold the stock **at least 1 trading day before the Book Closure Date** to be eligible for dividends, bonus shares, or rights. Buying on the Book Closure Date itself disqualifies you.

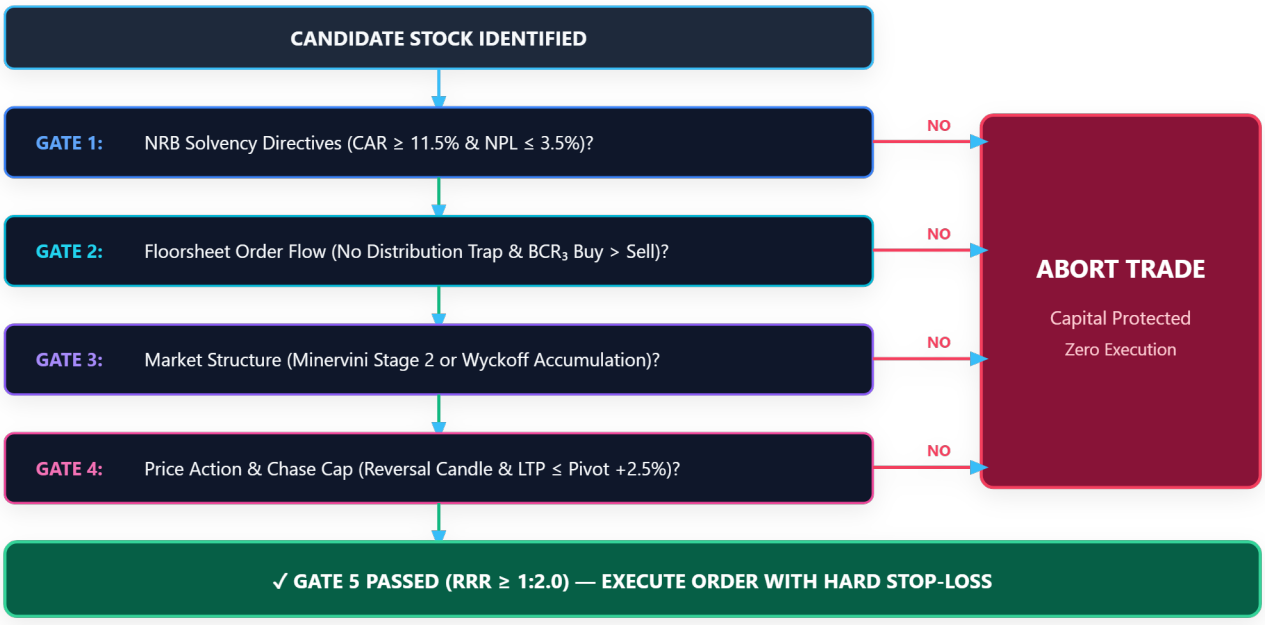
12.2 Quarter-End Window Dressing

- Occurs in **Ashoj, Poush, Chaitra, and Ashadh**.
- BFIs and mutual funds frequently purchase select benchmark stocks during the final 5 trading days of the quarter to artificially inflate their balance sheet asset valuations before publishing quarterly reports.

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13. Master Investor Decision Flowchart & Pre-Trade Checklist

FIGURE 13.1 — MASTER INVESTOR DECISION MATRIX & TRADE GATEKEEPER



The 5-Point Mandatory Pre-Trade Checklist:

1. **Fundamental Solvency**: Is EPS positive and NPL $\leq 3.5\%$?
2. **Promoter Supply Safety**: Is the stock clear of 3-year lock-in dump risk?
3. **Smart Money Clearance**: Is the Distribution Trap flag **FALSE** and BCR_3 $Buy > Sell$?
4. **Discipline at Pivot**: Is price strictly below the **Chase Cap ($Pivot + 2.5\%$)**?

5. **Execution Math**: Is the Risk-to-Reward Ratio at least $1:2.0$, with Target 1 defined and a non-negotiable Hard Stop-Loss placed?

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14. Empirical Discoveries, Edge-Case Audits & Real Case Studies

This section documents the specific empirical findings, mathematical edge-cases, and live exchange investigations audited directly from the Drabyashree NEPSE engine.

14.1 The $BCR_3 > 40|\%$ Paradox (The JOSHI / BEDC Investigation)

- **The Confusion**: Why does a stock with high buying concentration ($BCR_3\ Buy \geq 45|\%$) get flagged by the radar as a 🚨 **DISTRIBUTION TRAP**?
- **The Mathematical Reality**:
- In classic retail technical analysis, people only look at the buyer side: "*Broker #58 bought 45% of the shares today, so institutions are buying!*"
- **The Hidden Half of the Tape**: What did the selling side look like?
  - If top brokers bought $45|\%$ of the buy volume, but the top 3 selling brokers dumped $58|\%$ **of the sell volume**, the net balance of smart money is **negative**.
  - If the top seller is executing large block orders ($>20|\%$ of session turnover) into small 10–50 kitta retail bids on a green day (+2% to +5%), smart money is using **retail excitement as exit liquidity**.
- **The Invariant Rule**: A high $BCR_3\ Buy$ is **invalidated** if $BCR_3\ Sell \geq 45|\%$ and exceeds buy concentration by $\geq 10|\%$.

14.2 Real-World Case Study: GHL (Genuine Institutional Cornering)

- **Session Date**: October 13, 2026
- **Exchange Tape Metrics**:
- **Total Trades**: 1,743 trades
- **Traded Volume**: 1,216,448 kitta
- **Session Turnover**: Rs. 36.48 Crores (Rs. 364,778,570.80)
- **Block Trades ($\geq 1,000$ kitta or $\geq 3\ Lakhs$)**: 374 trades
- **Broker Concentration**:
- $BCR_3\ Buy = 41.0|\%$ (Broker #58: 21.9%, Broker #62: 9.9%, Broker #35: 9.1%)
- $BCR_3\ Sell = 25.5|\%$ (Broker #58: 14.1%, Broker #49: 6.0%, Broker #62: 5.3%)
- **Net Smart Money Flow**:
- **Broker #58**: Bought 266,668 kitta | Sold 171,477 kitta $|rightarrow$ **Net $+95,191$ kitta (+Rs. 2.85 Cr)**
- **Broker #62**: Net $+56,113$ kitta
- **Broker #35**: Net $+54,649$ kitta
- **Audit Verdict**:
- Buy concentration ($41.0|\%$) heavily dominates sell concentration ($25.5|\%$) by $+15.5|\%$.

- Top sellers are fragmented retail accounts (highest net seller was Broker #49 with only $-41k$ kitta).
- Verdict: 🟢 **INSTITUTIONAL ACCUMULATION CONFIRMED (Zero Trap)**.

14.3 Real-World Case Study: SADBL (The Complete Session Audit)

- **Session Date**: October 13, 2026
- **Exchange Tape Metrics**:
- **Total Trades**: Exactly **87 trades** (100% matched to NEPSE NOTS and Chukul feeds)
- **Traded Volume**: Exactly **35,119 kitta**
- **Session Turnover**: Exactly **Rs. 1,41,46,784.40** (accurate to the exact paisa)
- **First Trade**: Contract `2026101303000186` (Buyer #29, Seller #32, 50 kitta @ Rs. 404.00)
- **Last Trade**: Contract `2026101303010699` (Buyer #32, Seller #60, 40 kitta @ Rs. 405.00)
- **Key Investor Takeaways**:
- **Trade Counts Vary by Turnover**: Highly active momentum stocks (like GHL) have $>1,500$ trades, while moderate-cap development banks (like SADBL) have 50–150 trades. 87 trades is the **100% complete session** for SADBL.
- **Zero Truncation**: When selecting `[ 100 ]` or `[ All ]`, all 87 trades fit cleanly on Page 1 without pagination truncation.

14.4 Micro-Sample Distortion & Anti-Skew Guards

- **The Vulnerability**: In thinly traded or pre-open sessions with fewer than 15 trades (e.g. 18 trades on a micro-cap), a single retail order of 500 shares can make one broker appear to have 100% buy concentration.
- **The Engine Guard**:
- The app enforces strict sample sufficiency gates:

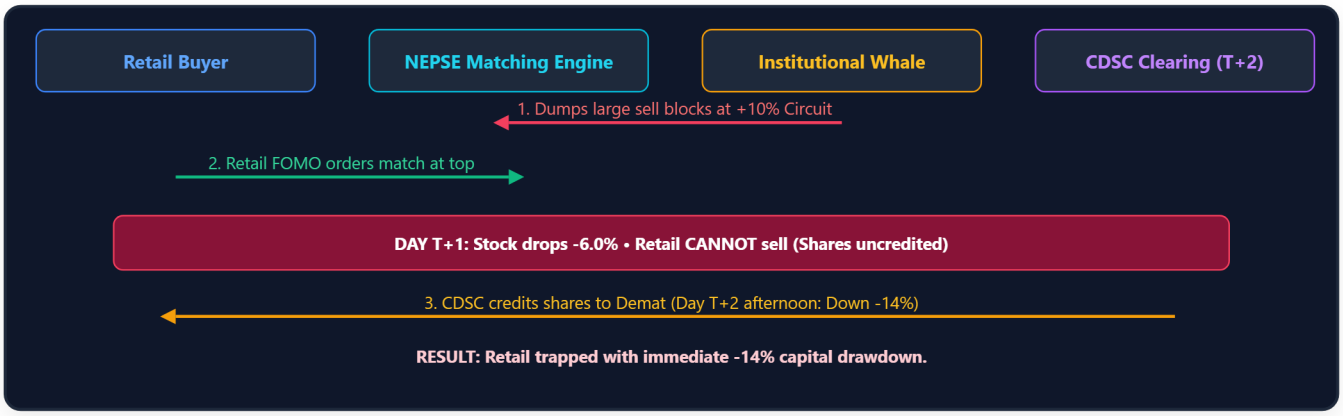
$$\text{Sample Valid} = \text{Trades} \geq 5 \quad \text{and} \quad \text{Buy Volume} \geq 200\text{–}500 \text{ kitta} \quad \text{or} \quad \text{Turnover} \geq \text{Rs. } 15 \text{ Lakhs}$$

- If a stock fails this threshold, concentration is forced to 0% and classified as **"Micro-Sample Dispersed Flow"** to prevent false institutional rankings.

14.5 The T+2 Settlement Trap Timeline

Understanding how an offloading trap destroys retail capital step-by-step:

FIGURE 14.1 — THE NEPSE T+2 SETTLEMENT TRAP SEQUENCE



14.6 Dual Floorsheet Zero-Sum Architecture

Every transaction executed on the Nepal Stock Exchange is a matched contract obeying the **Conservation of Volume Invariant**:

$$\sum \text{Buy Quantity} = \sum \text{Sell Quantity} = \text{Total Traded Quantity}$$

- In the app's Floorsheet table:
- **SYM**: Traded scrip ticker.
- **BB (Buyer Broker)**: Rendered in **Emerald Green** (e.g., `#58`).
- **SB (Seller Broker)**: Rendered in **Rose Red** (e.g., `#45`).
- **QTY & RATE**: Exact executed volume and rate in NPR.
- **AMOUNT**: Total rupee value of the contract.
- **CSV Export**: The `⬇ CSV` button exports 100% of the session's records with genuine numeric types, allowing investors to verify broker accumulation directly in Microsoft Excel.